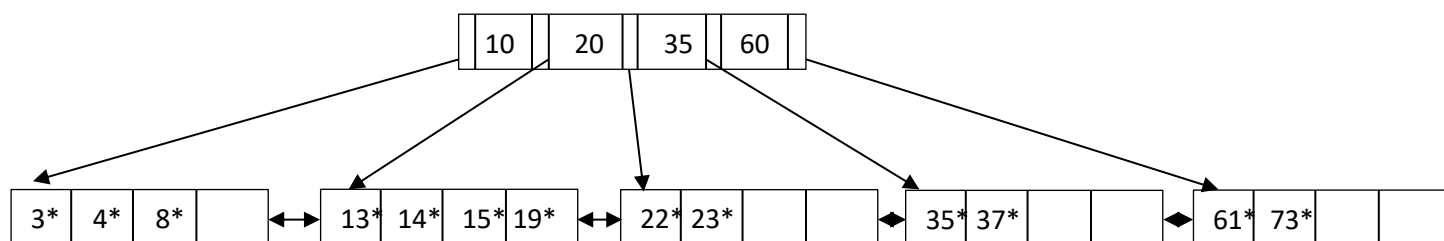


# CEN 301 Data Management and File Structures Midterm Exam

## QUESTIONS

*(There are 4 questions in this exam, you should submit until 14:00 to get full points)*

1. **(25 points)** Consider a disk with a sector size of 900 bytes, 800 tracks per surface, 120 sectors per track, three single-sided platters, and average seek time of 10 milliseconds. Disk platters rotate at 40 rps (revolutions per second), and block size is 2700 bytes. Suppose that a file containing 200000 records of 350 bytes each is to be stored on such disk and that no record is allowed to span two blocks.
  - a) Compute the capacity of a cylinder.
  - b) How many cylinders are required to store the file? Why?
  - c) Compute the average rotational delay.
  - d) Compute the block transfer time.
  
2. **(25 points)** You are given a file A that has 100000 employee records and each record occupies 350 bytes. The file A is stored over a disk with a block size of 1100 bytes and no record is allowed to span two blocks. Assume that we need to find the employees whose name start with letter 'B'. You may also assume that 10% of the employee records satisfy the condition given in the query. Calculate a timing estimate for disk I/O operations (disk accesses) to find the query result for each of the below file structures. You should ignore time required to write the query result, and explain your search algorithm in a few sentences. (Use  $s$  (seek time) = 16 milliseconds,  $r$  (average rotational delay) = 8.3 milliseconds,  $btt$  (block transfer time) = 0.84 milliseconds, you can use at most 2 buffer pages)
  - a) File A is a pile file.
  - b) File A is an indexed sequential file so that a B+ tree index for the employee name attribute has been created on file A. All the non-leaf nodes of the B+ tree are kept in memory. In the leaf nodes, the employee records are stored.
  
3. **(25 points)** Consider the B+ tree index structure given in the below figure:



- a) Draw the tree that would result from inserting data entry with key 16 into the above tree.
- b) Draw the tree that would result from deleting data entry with key 61 from the above tree.

4. (25 points) Insert the records with the keys **6, 3, 1, 11** into a hash table using **linear hashing**. Use  $h(\text{key}) = \text{key}$  as the hash function. Assume that initially, the buckets are empty, there are 2 data buckets ( $N=2$ ), each data page can hold at most 2 records,  $\text{Next}=0$ , and  $\text{Level}=0$ . Show each step of the insertion.

The initial hash table is given as follows:

$N=2, \text{level}=0, \text{Next}=0$

|       |                                                                                                                                                                                    |  |  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| $h_0$ |                                                                                                                                                                                    |  |  |
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